Manan Kochhar

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Education

Chitkara University — Chandigarh, India
Bachelor of Engineering — Computer Science Engineering, AI/ML Specialisation — *2025 – 2029*
- Relevant coursework: Machine Learning, Data Structures & Algorithms, Operating Systems, DBMS, Cloud Computing

Experience

AI Engineer — Kinetic City — Chandigarh, India
Finvasia Innovation Hackathon 2026 (Evolve-i) · Team Finbros · Track-1 PS-3 — *Jan 2026*
- Built an AI companion for first-time Indian retail investors, aimed at the fear that stops people investing at all
- Implemented Monte Carlo SIP portfolio simulation with rupee-framed output to cut decision paralysis
- Measured the result against investor confidence rather than model accuracy

Speech-to-Text Pipeline Lead — ASHA-VANI — Remote
Hinglish Voice AI Pipeline · Team Project — *2025*
- Led the end-to-end STT pipeline for ASHA health workers using Faster-Whisper and Silero VAD on an RTX 40-series GPU
- Added Hinglish and Bhojpuri support; benchmarked word error rate with `jiwer` across real-world audio conditions
- Authored a contract spec fixing 16 kHz mono int16 PCM so four developers could integrate against one audio format

Projects

Dhan Drishti | *TypeScript, React, Fastify, SQLite, Drizzle* — GitHub
- Self-hosted multi-broker portfolio tracker where every figure on the dashboard derives from a normalised transaction ledger
- Exact-decimal money end to end behind a pure, deterministic calculation engine covered by 352 tests; imports across 7 sources are idempotent, atomic and reconciled against each broker's own P&L report

Voice Emotion Detection | *Python, PyTorch, FastAPI, WebSockets* — GitHub
- Real-time speech emotion recognition over a four-model ensemble, with a neural VAD segmenting audio into complete utterances
- Built a two-corpus evaluation harness and used it to measure and drop several additions the numbers did not support; sub-second latency, analysed locally and never recorded

Virtual Keyboard & Air Mouse | *Python, OpenCV, MediaPipe, macOS Quartz* — GitHub
- Webcam-only touchless input: per-hand role assignment drives a 40-key on-screen keyboard and the real OS cursor
- Injects keystrokes and clicks at the operating-system level, so input lands in whatever application is focused; renders as a click-through overlay on macOS

Muse. v2 | *TypeScript, React, Socket.io, Prisma, PostgreSQL* — GitHub · Live
- Social listening platform whose rooms stay in sync from a single socket event carrying track and start time, rather than streaming audio between clients
- Content-based recommender weighting four taste signals into a per-user profile; canvas turntable with drag-to-scrub seeking

Synora | *JavaScript, React, Express, MySQL* — GitHub · Live
- Student workspace unifying tasks, timetable, attendance, CGPA and notes on a 20-table MySQL schema built around the subject list
- Seven runtime dependencies in total; the API self-migrates, applying missing tables and columns on its first request after a deploy

Certifications

Microsoft Certified: Azure AI Fundamentals (AI-900)	Microsoft · 2025
Develop Generative AI Applications	IBM · 2026
Build RAG Applications	IBM · 2026
Introduction to Generative AI Learning Path	Google · 2026

Technical Skills

Languages: Python, C, C++, JavaScript, TypeScript, SQL
AI & ML: PyTorch, TensorFlow, Keras, Scikit-learn, Hugging Face, LangChain, Faster-Whisper, Silero VAD
Vision & Speech: OpenCV, MediaPipe, Whisper, NumPy, Pandas
Web: React, Next.js, Node.js, FastAPI, Flask, Express, Tailwind CSS, Vite
Data, Cloud & Tools: PostgreSQL, MySQL, SQLite, Microsoft Azure, Docker, Git, NVIDIA CUDA